All Lie Jacobians

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1 Right vs. left Lie group Jacobians

For any non-commutative group, results differ depending on the order of the operands. This means that Jacobians can be defined in two ways. For manifolds $\mathcal{M}$ and $\mathcal{N}$ and a function $f : \mathcal{M} \to \mathcal{N}; \mathcal{X} \mapsto \mathcal{Y} = f(\mathcal{X})$, we have

Right Jacobian: using right plus in $\mathcal{M}$ and right minus in $\mathcal{N}$

$$\frac{{}^{\mathcal{X}}Df(\mathcal{X})}{D\mathcal{X}} \triangleq \lim_{\boldsymbol{\tau}\to 0} \frac{f(\mathcal{X} \oplus \boldsymbol{\tau}) \ominus f(\mathcal{X})}{\boldsymbol{\tau}} \triangleq \lim_{\boldsymbol{\tau}\to 0} \frac{\mathrm{Log}(f(\mathcal{X})^{-1} \circ f(\mathcal{X} \circ \mathrm{Exp}(\boldsymbol{\tau})))}{\boldsymbol{\tau}} \tag{1}$$

Left Jacobian: using left plus in $\mathcal{M}$ and left minus in $\mathcal{N}$

$$\frac{{}^{\mathcal{E}}Df(\mathcal{X})}{D\mathcal{X}} \triangleq \lim_{\boldsymbol{\tau}\to 0} \frac{f(\boldsymbol{\tau} \oplus \mathcal{X}) \ominus f(\mathcal{X})}{\boldsymbol{\tau}} \triangleq \lim_{\boldsymbol{\tau}\to 0} \frac{\mathrm{Log}(f(\mathrm{Exp}(\boldsymbol{\tau}) \circ \mathcal{X}) \circ f(\mathcal{X})^{-1})}{\boldsymbol{\tau}} \tag{2}$$

These two Jacobians work differently:

- The right Jacobian relates local perturbations of $\mathcal{X} \oplus \boldsymbol{\tau}$ around $\mathcal{X} \in \mathcal{M}$ with local perturbations of $f(\mathcal{X}) \oplus \mathbf{J}\boldsymbol{\tau}$ around $f(\mathcal{X}) \in \mathcal{N}$

$$f(\mathcal{X} \oplus {}^{\mathcal{X}}\boldsymbol{\tau}) \approx f(\mathcal{X}) \oplus \Big(\frac{{}^{\mathcal{X}}Df(\mathcal{X})}{D\mathcal{X}}\, {}^{\mathcal{X}}\boldsymbol{\tau}\Big) = \mathcal{Y} \oplus {}^{\mathcal{Y}}\boldsymbol{\sigma} \tag{3}$$

- The left Jacobian relates global perturbations of $\boldsymbol{\tau} \oplus \mathcal{X}$ around the origin $\mathcal{E} \in \mathcal{M}$ with local perturbations of $\mathbf{J}\boldsymbol{\tau} \oplus f(\mathcal{X})$ around the origin $\mathcal{E} \in \mathcal{N}$.

$$f({}^{\mathcal{E}}\boldsymbol{\tau} \oplus \mathcal{X}) \approx \Big(\frac{{}^{\mathcal{E}}Df(\mathcal{X})}{D\mathcal{X}}\, {}^{\mathcal{E}}\boldsymbol{\tau}\Big) \oplus f(\mathcal{X}) = {}^{\mathcal{E}}\boldsymbol{\sigma} \oplus \mathcal{Y} \tag{4}$$

Since local and global perturbations are related by the adjoint (**??**), that is,

$$ {}^{\mathcal{E}}\boldsymbol{\tau} = \mathbf{Ad}_{\mathcal{X}}\, {}^{\mathcal{X}}\boldsymbol{\tau} \qquad\qquad {}^{\mathcal{E}}\boldsymbol{\sigma} = \mathbf{Ad}_{\mathcal{Y}}\, {}^{\mathcal{Y}}\boldsymbol{\sigma} \tag{5}$$

we have that,

$$\frac{{}^{\mathcal{E}}Df(\mathcal{X})}{D\mathcal{X}} \mathbf{Ad}_{\mathcal{X}} = \mathbf{Ad}_{f(\mathcal{X})} \frac{{}^{\mathcal{X}}Df(\mathcal{X})}{D\mathcal{X}}\ . \tag{6}$$

Expressions for the left and right Jacobians are given in Table 1 as functions of $\mathbf{J}_l$, $\mathbf{J}_r$ and $\mathbf{Ad}$, which must be determined for each particular group. We have also the following binary,

Table 1: Left and right Jacobians of Lie groups in terms of $\mathbf{J}_l$, $\mathbf{J}_r$ and $\mathbf{Ad}$.

Oper.	Left Jacobians		Right Jacobians	
$\mathcal{X}^{-1}$	$\mathbf{J}_{\mathcal{X}}^{\mathcal{X}^{-1}} = -\mathbf{Ad}_{\mathcal{X}}{}^{-1}$		$\mathbf{J}_{\mathcal{X}}^{\mathcal{X}^{-1}} = -\mathbf{Ad}_{\mathcal{X}}$	
$\mathcal{X} \circ \mathcal{Y}$	$\mathbf{J}_{\mathcal{X}}^{\mathcal{X}\circ\mathcal{Y}} = \mathbf{I}$	$\mathbf{J}_{\mathcal{Y}}^{\mathcal{X}\circ\mathcal{Y}} = \mathbf{Ad}_{\mathcal{X}}$	$\mathbf{J}_{\mathcal{X}}^{\mathcal{X}\circ\mathcal{Y}} = \mathbf{Ad}_{\mathcal{Y}}^{-1}$	$\mathbf{J}_{\mathcal{Y}}^{\mathcal{X}\circ\mathcal{Y}} = \mathbf{I}$
$\mathrm{Exp}(\boldsymbol{\tau})$	$\mathbf{J}_{\boldsymbol{\tau}}^{\mathrm{Exp}(\boldsymbol{\tau})} = \mathbf{J}_l(\boldsymbol{\tau})$		$\mathbf{J}_{\boldsymbol{\tau}}^{\mathrm{Exp}(\boldsymbol{\tau})} = \mathbf{J}_r(\boldsymbol{\tau})$	
$\mathrm{Log}(\mathcal{X})$	$\mathbf{J}_{\mathcal{X}}^{\mathrm{Log}(\mathcal{X})} = \mathbf{J}_l^{-1}(\boldsymbol{\tau})$		$\mathbf{J}_{\mathcal{X}}^{\mathrm{Log}(\mathcal{X})} = \mathbf{J}_r^{-1}(\boldsymbol{\tau})$	
Left Plus	$\mathbf{J}_{\mathcal{X}}^{\boldsymbol{\tau}\oplus\mathcal{X}} = \mathbf{Ad}_{\mathrm{Exp}(\boldsymbol{\tau})}$	$\mathbf{J}_{\boldsymbol{\tau}}^{\boldsymbol{\tau}\oplus\mathcal{X}} = \mathbf{J}_l(\boldsymbol{\tau})$	$\mathbf{J}_{\mathcal{X}}^{\boldsymbol{\tau}\oplus\mathcal{X}} = \mathbf{I}$	$\mathbf{J}_{\boldsymbol{\tau}}^{\boldsymbol{\tau}\oplus\mathcal{X}} = \mathbf{Ad}_{\mathcal{X}}{}^{-1}\mathbf{J}_r(\boldsymbol{\tau})$
Left Minus	$\mathbf{J}_{\mathcal{X}}^{\mathcal{X}\ominus\mathcal{Y}} = \mathbf{J}_l^{-1}(\boldsymbol{\tau})$	$\mathbf{J}_{\mathcal{Y}}^{\mathcal{X}\ominus\mathcal{Y}} = -\mathbf{J}_r^{-1}(\boldsymbol{\tau})$	$\mathbf{J}_{\mathcal{X}}^{\mathcal{X}\ominus\mathcal{Y}} = \mathbf{J}_r^{-1}(\boldsymbol{\tau})\mathbf{Ad}_{\mathcal{Y}}$	$\mathbf{J}_{\mathcal{Y}}^{\mathcal{X}\ominus\mathcal{Y}} = -\mathbf{J}_r^{-1}(\boldsymbol{\tau})\mathbf{Ad}_{\mathcal{Y}}$
Right Plus	$\mathbf{J}_{\mathcal{X}}^{\mathcal{X}\oplus\boldsymbol{\tau}} = \mathbf{I}$	$\mathbf{J}_{\boldsymbol{\tau}}^{\mathcal{X}\oplus\boldsymbol{\tau}} = \mathbf{Ad}_{\mathcal{X}}\mathbf{J}_l(\boldsymbol{\tau})$	$\mathbf{J}_{\mathcal{X}}^{\mathcal{X}\oplus\boldsymbol{\tau}} = \mathbf{Ad}_{\mathrm{Exp}(\boldsymbol{\tau})}^{-1}$	$\mathbf{J}_{\boldsymbol{\tau}}^{\mathcal{X}\oplus\boldsymbol{\tau}} = \mathbf{J}_r(\boldsymbol{\tau})$
Right Minus	$\mathbf{J}_{\mathcal{X}}^{\mathcal{X}\ominus\mathcal{Y}} = \mathbf{J}_l^{-1}(\boldsymbol{\tau})\mathbf{Ad}_{\mathcal{Y}}{}^{-1}$	$\mathbf{J}_{\mathcal{Y}}^{\mathcal{X}\ominus\mathcal{Y}} = -\mathbf{J}_l^{-1}(\boldsymbol{\tau})\mathbf{Ad}_{\mathcal{Y}}{}^{-1}$	$\mathbf{J}_{\mathcal{X}}^{\mathcal{X}\ominus\mathcal{Y}} = \mathbf{J}_r^{-1}(\boldsymbol{\tau})$	$\mathbf{J}_{\mathcal{Y}}^{\mathcal{X}\ominus\mathcal{Y}} = -\mathbf{J}_l^{-1}(\boldsymbol{\tau})$

ternary and quaternary identities [1]

$$\mathbf{J}_l(\boldsymbol{\tau}) = \mathbf{J}_r(-\boldsymbol{\tau}) \tag{7}$$

$$\mathbf{J}_l(\boldsymbol{\tau}) = \mathbf{Ad}_{\mathrm{Exp}(\boldsymbol{\tau})}\mathbf{J}_r(\boldsymbol{\tau}) \tag{8}$$

$$\frac{{}^{\mathcal{E}}D\mathcal{Y}}{D\mathcal{X}}\mathbf{Ad}_{\mathcal{X}} = \mathbf{Ad}_{\mathcal{Y}}\frac{{}^{\mathcal{X}}D\mathcal{Y}}{D\mathcal{X}} \tag{9}$$

References

[1] J. Solà, J. Deray, and D. Atchuthan, “A micro Lie theory for state estimation in robotics,” Institut de Robòtica i Informàtica Industrial, Barcelona, Tech. Rep. IRI-TR-18-01, 2018.